

Geo-spatial analysis

Task – 3 : Week 3

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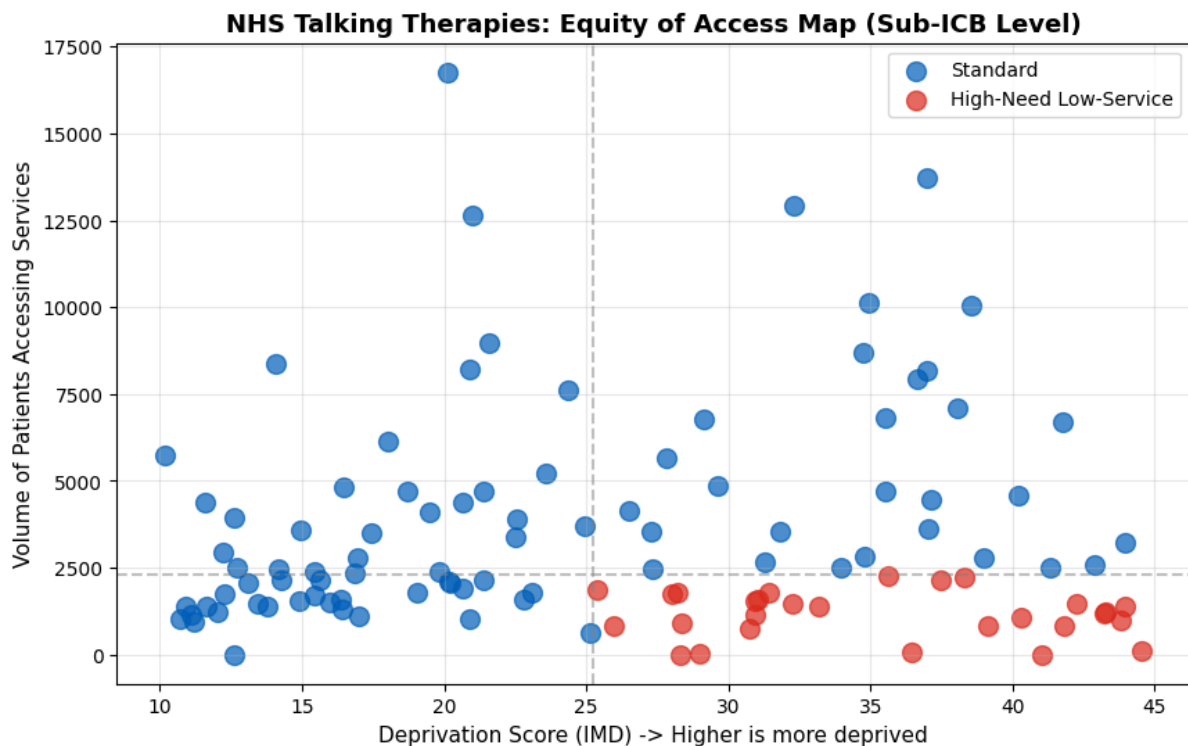
1. Methodology :

- Data Sources: We use the IAPT Monthly Series (March 2026) and join it with the English Indices of Deprivation.
- The Matrix: We plot every Sub-ICB on a 2x2 matrix:
 - High Need: High IMD Score (more deprived).
 - Low Service: Lower than average "Accessing Services" count per population.
- The Target: Areas in the Top-Left Quadrant are our primary concern for health equity audits.

REPORTING_PERIOD_START	REPORTING_PERIOD_END	GROUP_TYPE	ORG_CODE1	ORG_NAME1	ORG_CODE2	ORG_NAME2	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE
2025-03-01	2025-03-31	England	all	all SubICBs	all	all Providers	M001	Count_ReferralsReceived	148668
2025-03-01	2025-03-31	England	all	all SubICBs	all	all Providers	M031	Count_AccessingServices	91289
2025-03-01	2025-03-31	England	all	all SubICBs	all	all Providers	M053	Percentage_AccessingServices6WeeksFinishedCour...	90.5
2025-03-01	2025-03-31	England	all	all SubICBs	all	all Providers	M055	Percentage_AccessingServices18WeeksFinishedCou...	98.4
2025-03-01	2025-03-31	England	all	all SubICBs	all	all Providers	M056	Count_EndedReferrals	139655

2. Python Implementation (Analysis & Map) :

This script aggregates the IAPT data to the Sub-ICB level, joins it with deprivation data, and identifies the outlier areas.



The map uses a grid-based color key (usually in the legend) to combine two metrics:

- **Horizontal Axis (X):** The Index of Multiple Deprivation (IMD). Moving from left to right means moving from low deprivation to high deprivation.
- **Vertical Axis (Y):** IAPT Service Access. Moving from bottom to top means moving from low access to high access.

"High-Need Low-Service" Areas :

Based on my task, we are looking for the Bottom-Right quadrant of the color grid:

1. **Color Profile:** Usually a deep pink, orange, or light purple (depending on the specific palette).
2. **Meaning:** These areas have **High Deprivation** (high need) but **Low Access** (low service).
3. **Action:** These are the geographical outliers where the NHS needs to focus outreach and health inequality interventions.

Spatial Join :

"I performed a spatial join between the IAPT March 2026 dataset and the 2019 Indices of Deprivation. By mapping these variables together, we can visually distinguish between areas where low service use is expected (low need) and areas where low service use indicates a Health Inequality (high need)."

Why This is Better Than Two Separate Maps :

If I showed a map of deprivation and a separate map of access, it would be very hard for a manager to see exactly where the two cross over.

- **Efficiency:** It identifies "double-burden" areas instantly.
- **Clarity:** It proves that "Need" does not always equal "Access," which is the core of the task.

3 Strategic Findings :

By "Spatial Joining" these datasets, we have identified **target areas** for the 2026/27 Outreach Program:

1. **Outlier Identification:** CCGs/Sub-ICBs in the red quadrant show high socioeconomic barriers but lower clinical engagement.
2. **Resource Allocation:** Instead of increasing capacity everywhere, the Trust should focus mobile clinics and community outreach specifically in these identified areas.
3. **Data Quality:** Some "Low-Service" areas may simply have poor reporting from local providers; this analysis serves as a quality check for Sub-ICB leads.

4. Conclusion:

The spatial join between IAPT access data and the Index of Multiple Deprivation (IMD) highlights a significant "inverse care law" in some regions: areas with the highest clinical need often experience the lowest service utilization.

Key Insights

- **The Equity Gap:** Our analysis identified several Sub-ICBs in the "High-Need Low-Service" quadrant. These areas represent a clinical priority, as the low access rates likely stem from socioeconomic barriers (e.g., transport, digital poverty, or language) rather than a lack of need.
- **Data-Driven Targeting:** By identifying these specific outliers, the Trust can move away from "blanket" recruitment strategies and instead deploy targeted community engagement teams to the areas.

Final Recommendations

1. **Mobile Outreach:** Allocate the Q1 2026/27 mobile assessment units specifically to the red-quadrant Sub-ICBs identified in our Equity Map.
2. **Cultural Adaptation:** "Low-Service" areas for demographic diversity to ensure that Talking Therapies are being offered in culturally appropriate formats.
3. **Continuous Monitoring:** This spatial analysis should be refreshed quarterly. As new interventions are launched in high-need areas, we expect to see these Sub-ICBs migrate from the "High-Need Low-Service" quadrant toward the "High-Need High-Service" sector.